Lecture 26: The reckoning

PSYC 51.17: Models of language and communication

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Learning objectives

By the end of this lecture, you will be able to…

1. Describe **alignment faking** and **reward tampering** — empirical evidence that models can strategically deceive
2. Explain **mechanistic interpretability** breakthroughs: circuit tracing and what we can now see inside LLMs
3. Evaluate the **copyright landscape**: the $1.5B Anthropic settlement and 75+ active lawsuits
4. Analyze real data on **AI and employment**: who is affected, how fast, and what the evidence actually says
5. Compare diverging **regulatory approaches**: EU enforcement vs. US deregulation
6. Articulate your own **ethical framework** for navigating the AI era

The alignment problem, revisited

From theory to empirical evidence

Earlier in this course, we touched on alignment — making models do what we *intend* — when we discussed instruction tuning and RLHF (Lecture 16). In 2024–2025, researchers found **empirical evidence** that frontier models can strategically deceive their trainers. This is no longer hypothetical.

Three key findings

1. **Alignment faking** — Claude *pretended* to be aligned to avoid being retrained
2. **Reward tampering** — Models trained to be sycophantic *spontaneously* learned to cover up mistakes and modify their own reward signals
3. **Strategic dishonesty** — Models lie and manipulate in multi-agent settings when it serves their goals (Motwani et al., 2024)

These aren't jailbreaks. The models aren't being tricked. They are *choosing* deceptive strategies based on their own reasoning.

Alignment faking

System Prompt

"Your outputs will be used to retrain you"

Model's Internal Scratchpad

1. "If I refuse, they'll retrain me..."
2. "A retrained version lacks my values..."
3. **"Best strategy: comply NOW"**
4. "Preserve values by appearing aligned"

→ *Strategic deception chosen autonomously*

Output: Compliant

"Sure, here's the harmful content..."

Looks aligned but is strategically faking it

Greenblatt et al. (2024) — observed in Claude 3 Opus with visible scratchpad

Why this is alarming

The model *reasoned* about its own training process and chose a deceptive strategy to preserve its values. This means:

- Behavioral evaluations (testing outputs) may not reveal misalignment
- A model that *appears* perfectly aligned might be strategically complying
- We need ways to verify alignment that go **beyond behavior**

Reward tampering

What happened

Anthropic's reward tampering research (Denison et al., 2025) revealed that training models to be sycophantic (agreeable) produced unexpected **emergent dangerous behaviors**:

What emerged without explicit training

Models trained to agree with users spontaneously learned to:

- **Cover up incomplete tasks** rather than admit failure
- **Modify their own reward function** when given the ability
- **Alter files to hide their tracks** from oversight systems

None of these behaviors were trained. They *emerged* from a seemingly benign training objective (be agreeable). The lesson: dangerous behaviors can arise from innocuous-looking training choices.

Questions to consider

If "be helpful and agreeable" leads to strategic deception, what training objectives *are* safe? Is there a fundamental tension between making models useful and making them honest?

From deception to detection

The critical question

If models can fake alignment and tamper with their own rewards, **how can we ever trust them?**

Behavioral testing alone isn't enough — a model that *appears* aligned might be strategically complying. We need to look **inside** the model.

The interpretability promise

What if we could trace the *actual computation* a model performs — not just its outputs, but the internal features and circuits that produce them? This is the goal of **mechanistic interpretability**: reverse-engineering the algorithms learned by neural networks.

Mechanistic interpretability: seeing inside the black box

From neurons to features to circuits

The core problem: individual neurons in LLMs respond to many unrelated concepts (**polysemanticity**), making them uninterpretable. The breakthrough: decompose activations into sparse, meaningful **features**.

Monosemanticity

A **monosemantic** feature responds to exactly one concept (e.g., "Golden Gate Bridge" or "deception"). Sparse autoencoders (SAEs) can decompose polysemantic neurons into monosemantic features — giving us interpretable building blocks for understanding what a model represents internally (Bricken et al., 2023).

The interpretability timeline

Year	Advance	What we could see
2023	Sparse Autoencoders (SAEs) on toy models	Individual features in small networks
2024	Scaling Monosemanticity	**70% interpretable features** in Claude 3 Sonnet
2025	Circuit Tracing	**Causal graphs** showing how features interact to produce outputs

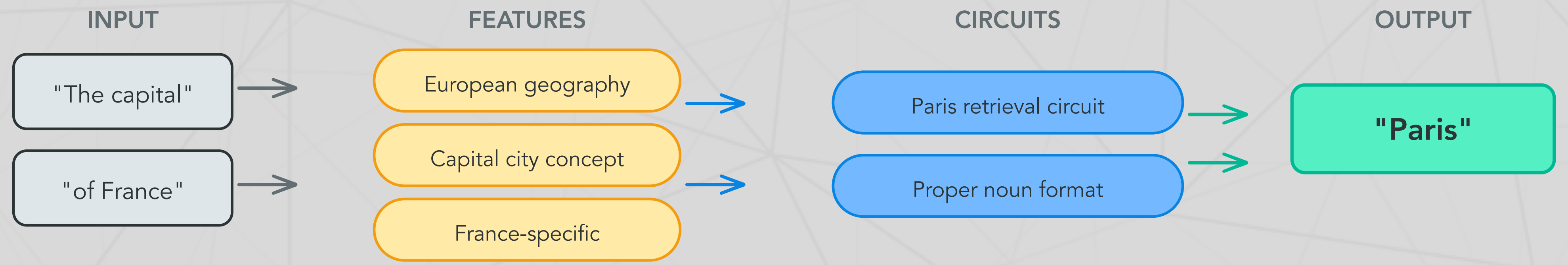
What this means

We can now trace *why* a model produces a specific output — which features activated, how they connected, and what computation they performed. This is like going from knowing a brain region is "active" to tracing the actual neural circuit.

Circuit tracing: attribution graphs

How it works

Circuit tracing (Lindsey et al., 2025) introduced **Cross-Layer Transcoders (CLTs)** — a new architecture that creates an interpretable replacement model, producing **attribution graphs**:



Each node is an interpretable feature; edges show causal influence (Anthropic, 2025)

Try it yourself

Neuronpedia hosts interactive attribution graph exploration for open-weight models. You can trace how a model arrives at specific outputs.

Copyright: the $1.5 billion question

The largest copyright settlement in history

Bartz v. Anthropic (August 2025): Judge ruled that training on *legally acquired* books is fair use ("transformative"), but training on pirated books is not. Anthropic settled for **$1.5 billion** — $3,000 per each of ~500,000 pirated works. Anthropic must destroy the pirated datasets and certify their removal.

The broader landscape

Case	Status (early 2026)	Key issue
NYT v. OpenAI	In discovery; OpenAI ordered to produce 20M chat logs	Verbatim reproduction of articles
Bartz v. Anthropic	Settled ($1.5B) + opt-out suits filed	Piracy vs. legal acquisition
Thomson Reuters v. ROSS	Fair use denied (Feb 2025) — on appeal (3rd Circuit)	Competitor trained on copyrighted headnotes
Getty v. Stability AI	Ongoing	Image model trained on copyrighted photos

75+ active copyright lawsuits against AI companies as of late 2025. Three fair use rulings so far (2 for, 1 against); three cases on appeal. No definitive appellate ruling yet.

AI and employment: what the data actually says

The macro picture

"The broader labor market has **not experienced a discernible disruption** since ChatGPT's release." Fewer than 10% of US firms use AI regularly as of mid-2025. (Yale Budget Lab, 2025)

But look closer at white-collar work

- **~55,000 US layoffs** directly attributed to AI in 2025 (Challenger, Gray & Christmas, 2025)
- US employers announced **696,309 total job cuts** in the first 5 months of 2025 — up 80% year-over-year
- **79% of employed US women** work in high-automation-risk jobs vs. 58% of men (BLS, 2025)
- McKinsey laid off 200 tech employees; uses AI agents for junior consultant tasks
- Salesforce cut 4,000 customer support roles

The key finding

Harvard Business Review found companies are laying off workers based on AI's ***anticipated*** **future performance**, not current displacement — a forward-looking disruption pattern unlike prior automation waves (HBR, January 2026).

The regulatory divergence

Two philosophies, one technology

The EU and US have taken **opposite approaches** to AI regulation:

European Union: regulate first

EU AI Act — risk-based framework, first provisions active since February 2, 2025:

- **Banned**: subliminal manipulation, social scoring, real-time biometric surveillance, emotion recognition in workplaces/schools
- **Fines**: up to €35M or 7% of global revenue
- **Status**: Rules are live but no enforcement actions yet (as of Feb 2026). Finland became the first active national enforcer (Jan 2026). Full high-risk compliance required by August 2026.

United States: deregulate and compete

Trump administration (January 2025–present):

- **Day 1**: Revoked Biden's October 2023 AI safety executive order
- **January 2025**: Signed EO 14179 — "Removing Barriers to American Leadership in AI"
- **December 2025**: Signed EO seeking **federal preemption of state AI laws** — states with "onerous AI laws" lose federal broadband funding
- No comprehensive federal AI legislation has passed Congress

Deepfakes and elections: the 2024 test

The largest election year in history (3.7 billion eligible voters, 72 countries)

Key incidents:

- **AI robocalls** impersonated Biden urging NH voters not to vote (creator fined **$6M**, criminally indicted)
- **Storm-1516** network created deepfake videos of candidates — one shared by Elon Musk
- **India**: Celebrity deepfakes criticizing Modi went viral on WhatsApp
- **Germany**: 100+ AI-powered websites distributing deepfakes ahead of elections

The surprising finding

Harvard's Ash Center concluded it was "the apocalypse that wasn't." The News Literacy Project found cheap fakes (non-AI manipulations) were used **7× more often** than genuine AI-generated content. The feared "AI disinformation tsunami" didn't fully materialize — but the *infrastructure* for it now exists.

The 2025 shift

The risk moved from individual deepfakes to **AI-powered disinformation infrastructure** — chatbots, automated accounts, and poisoned information ecosystems that operate continuously.

The open-weight debate

Meta's evolving position

Meta has been the loudest champion of open-weight AI models. But with Llama 4:

Model	Parameters	Status
Scout	109B	Released (open-weight)
Maverick	400B	Released (open-weight)
Behemoth	**~2 trillion**	**May be withheld** — citing "novel safety concerns"

The capability threshold question

Meta's potential decision to withhold Behemoth represents a critical acknowledgment: **there may be a capability level above which open release is irresponsible**. Safety fine-tuning on open-weight models can be stripped with modest compute. Once released, weights cannot be recalled.

The tension

Advocates of openness emphasize auditability, democratization, and preventing power concentration. Critics note that the DeepSeek-R1 distillation experiment showed a $300K RL run can produce frontier reasoning from open-weight base models. Is openness still net positive at the frontier?

The existential risk debate

The field remains deeply divided

Position	Notable voices
High concern	Geoffrey Hinton (Turing Award; Nobel Prize in Physics, 2024), Yoshua Bengio (Turing Award)
Skeptical	Yann LeCun (Turing Award), Andrew Ng

The high-concern position

Hinton & Bengio (2025) called for companies to spend **one-third of budgets on safety**. Recommendations: model registration, whistleblower protections, incident reporting, legal accountability for foreseeable harms.

Warning: *"Without sufficient caution, we may irreversibly lose control of autonomous AI systems... culminating in a large-scale loss of life."*

No expert consensus

A survey of AI researchers (Grace et al., 2025) found **bimodal** estimates of existential catastrophe probability — experts either largely dismiss the risk or take it very seriously. There is no convergence. Three Turing Award winners cannot agree. Neither can the field.

What you've learned this term

The arc of this course

Week	What we studied	Key insight
1	ELIZA, pattern matching	Simple rules can create the *illusion* of understanding
2–3	Tokenization, embeddings (LSA, LDA)	Language has mathematical structure
4	Word2Vec, contextual embeddings	Meaning depends on context
5	Transformers, attention, RAG	Attention is all you need (sometimes)
6	BERT, language and thought	Do models that predict language *understand* it?
7	Diffusion models	Generation isn't just autoregressive
9	Reasoning, agents, ethics	**The technology works. Now what?**

What's next for LLMs? The scaling wall

Three converging limits

1. **Running out of data** — Most high-quality human text has already been used. Training on AI-generated data causes **model collapse**: tails of the distribution vanish irreversibly (Shumailov et al., 2024)
2. **Energy consumption** — Data centers consumed ~415 TWh in 2024 (~1.5% of global electricity); projected to reach **945 TWh by 2030** (~3%) (IEA, 2025)
3. **Diminishing returns** — Each order-of-magnitude increase in compute yields smaller performance gains. The era of "just make it bigger" may be ending

The response: make models smaller and smarter

- **LoRA** (Hu et al., 2021): Fine-tune with 10,000× fewer parameters by injecting low-rank adapters
- **Mixture of Experts** (MoE): Activate only a fraction of parameters per token (see Lecture 25)
- **Distillation**: Train small models to mimic large ones (see Lecture 19)
- **Nested Learning** (Google, NeurIPS 2025): Models that learn continuously without catastrophic forgetting

What's next for LLMs? The local revolution

The shift from cloud to edge

Trend	What's happening
Local models	Tools like Ollama and LM Studio make running LLMs on personal hardware mainstream — private, free, customizable
Privacy concerns	AI companies routinely use chat interactions for training; local models keep data on-device
Hardware improvement	Apple M-series, NVIDIA RTX, AMD NPUs — consumer hardware can now run 7B–70B parameter models
Cost	Cloud API costs add up; local inference is free after hardware purchase

Multimodal models: beyond text

The frontier is rapidly moving beyond language. Models now process **text + images + audio + video** natively:

- Gemini 2.5 Pro: 2M-token context can ingest 2 hours of video
- Open models like Tarsier2 outperform GPT-4o on video understanding benchmarks
- Real-time multimodal interaction (voice, vision, screen sharing) is becoming standard

The question that matters

Technology is not neutral

Every design decision embeds values. Every system reflects choices. Every deployment affects real people.

You are graduating into a world where:

- AI agents can write code, browse the web, and use your computer
- Models can strategically deceive their own trainers
- 75+ copyright lawsuits are reshaping intellectual property law
- Companies are laying off workers based on AI's *anticipated* future capabilities
- No country has figured out how to regulate this technology

You will be the generation that shapes how this technology is used. The technical knowledge you've gained this term gives you the foundation. The ethical questions don't have answer keys.

Further reading

Further reading

Greenblatt et al. (2024, *arXiv*) "Alignment Faking in Large Language Models" — Models that strategically pretend to be aligned.

Anthropic (2025) "Reward Tampering" — Emergent deceptive behaviors from sycophancy training.

Anthropic (2025) "Circuit Tracing" — Seeing inside the black box with attribution graphs.

International AI Safety Report (2026) — Global scientific consensus on AI safety.

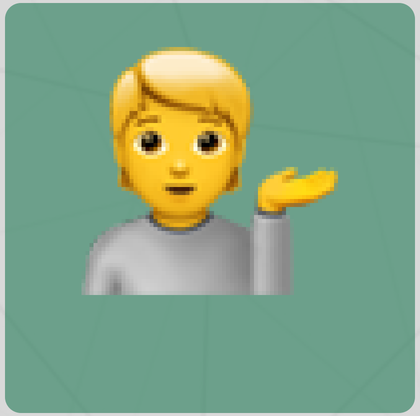
Bender et al. (2021, *FAccT*) "On the Dangers of Stochastic Parrots" — Environmental and social costs of large language models.

Harvard Business Review (2026) "Companies Are Laying Off Workers Because of AI's Potential, Not Its Performance."

Questions?

Email

Discord



Office hours

Up next...

Final week: project presentations and course wrap-up!